TIM ZHANG

AI Engineer | Agentic & RAG Systems | Data Architect

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SUMMARY

MSc candidate in Artificial Intelligence and Business Analytics with hands-on experience designing production-grade Agentic and RAG systems. Skilled in end-to-end AI engineering across multi-agent orchestration, semantic retrieval (pgvector + RRF), LLM-as-Judge quality evaluation, and cloud-native deployment. Comfortable bridging research prototypes with deployable services on Google Cloud Run, FastAPI, and PostgreSQL.

EDUCATION

Lingnan University, Hong Kong SAR *09/2025 – 08/2026 (Expected)*

MSc in Artificial Intelligence and Business Analytics

- **Key Coursework:** Foundations of AI, Deep Learning, Large Language Models, Blockchain, Business Data Management

Shenzhen University, China *09/2013 – 06/2017*

Bachelor of Management

KEY PROJECTS

SRR Agentic Case Processing System · Technical Lead *10/2025 – 02/2026*

Lingnan Cup Competition Project · Partner: HK Architectural Services Department (ArchSD) · github.com/SRR-Project-Team/SRR-Agentic-Case-Processing-System

- Designed a 7-layer Agentic processing architecture composed of 17 pluggable atomic capabilities, enabling end-to-end automation of ICC 1823, TMO, and RCC case channels.
- Built multi-format document parsing pipeline with field extraction across A–Q categories; integrated OpenAI GPT-4o + text-embedding-3-small for classification and response generation.
- Implemented pgvector-based semantic case retrieval with Reciprocal Rank Fusion (RRF) ranking, and a 3-tier quality assessment system (L1 keyword → L2 rules → L3 RAGAS LLM-as-Judge).
- Engineered a self-healing mechanism with Best-of-N selection and 4-category failure differential rollback; deployed on Google Cloud Run + Cloud SQL with Docker for auto-scaling.
- **Results:** 67 commits and 80,430 lines of code across two iterations; 12/12 requirements fully delivered; processing time reduced by 75% (40 min → 10 min); projected annual cost savings of HK$6.39M.

Tech Stack: Python 3.11, FastAPI, React + TypeScript, PostgreSQL 15 + pgvector, OpenAI GPT-4o, Google Cloud Run, Docker

RAGagent — Lingnan University Policy Q&A System · Developer *01/2026 – Present*

github.com/terzotian/RAGagent

- Built a hybrid semantic search pipeline combining pgvector embeddings and TF-IDF scoring, fused via Reciprocal Rank Fusion (RRF) for higher retrieval precision.
- Integrated Google Vertex AI (Gemini) with text-embedding-004 for generative responses; implemented streaming output for real-time answer generation.
- Delivered user management, file upload, and conversation history features for a production-ready deployment.
- **Results:** Policy Q&A accuracy >95%; average retrieval time reduced by >70%; 37 commits, 86.3% Python codebase.

Tech Stack: React + TypeScript + Vite, FastAPI, PostgreSQL + pgvector, Google Vertex AI (Gemini)

Cloud-Native Enterprise Data Warehouse Modernization · Lead Architect *2023 – 2025*

End-to-end redesign of a 10TB+ data platform for AI/BI workloads

- Restructured a legacy on-premise warehouse into a 5-layer ODS / DIM / DWD / DWS / ADS architecture on Tencent Cloud, optimizing data flow and downstream feature consumption.

- Implemented Seatunnel-based incremental and full data syncs, achieving 99.95% data consistency and reducing sync failures by 70%.
- Optimized storage with ORC format and partitioning/bucketing strategies; integrated StarRocks with Hive external tables for sub-second analytical queries.
- **Results:** 10TB+ historical data migrated; 87.5% ETL time reduction (hours → minutes); 38% storage savings; average query latency 0.8s (down from 3–10 minutes); 99.7% scheduling on-time rate.

Tech Stack: Python, SparkSQL, StarRocks, Hive, DolphinScheduler, Seatunnel, Tencent Cloud

B2B Cross-Source Data Quality Monitoring Platform · Data Engineer *2023 – 2025*

Automated anomaly detection and reconciliation across heterogeneous financial systems

- Built cross-source financial data quality monitoring with automated anomaly detection rules (e.g., fulfillment amount exceeding order amount) over 414,000+ contract records.
- Designed B2B reporting iterations covering contract lifecycle, countdown tracking, and credit ledger reconciliation across StarRocks and SQL Server.
- Implemented Feishu-based alerting for downstream BI consumers, reducing data incident detection latency from days to minutes.

Tech Stack: Python, SQL, StarRocks, SQL Server, DolphinScheduler, Feishu API, Power BI

TECHNICAL SKILLS

AI / ML Engineering: RAG Architecture, Agentic Workflows, LLM-as-Judge (RAGAS), Prompt Engineering, OpenAI GPT-4o, Google Vertex AI (Gemini), LangChain, FastAPI, pgvector

Programming: Python, SQL (BigQuery / SparkSQL / Impala), Bash / Shell, TypeScript

Data & Big Data: Hadoop, Apache Spark, Apache Hive, StarRocks, Kafka, PostgreSQL, MySQL, MongoDB, DBT

Cloud & DevOps: Google Cloud Platform (Cloud Run, Cloud SQL), Tencent Cloud, Docker, Linux

Languages: Mandarin (Native), Cantonese (Native), English (Professional Working Proficiency)